WORKED SOLUTION

STEP II 2014, Question 2

Pure • Integral Inequalities

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q2

This question has a full guided treatment in *Beyond STEP*, Section 3.7, $\pi^2 \leq 10$: https://beyondstep.keithmcnulty.org/ch03/wirtinger.html

The key idea

Both sides of $(*)$ are integrals we can work out exactly for any explicit f, so every function that vanishes at both ends turns $(*)$ into an inequality between numbers involving π. The only real choice is to pick the free constants so that $f(0) = f(\pi) = 0$. A function that does *not* vanish at an end can be large while changing slowly, which is why the end conditions matter.

Solution

Part (i)

With $f(x) = \sin nx$ we have $f'(x) = n\cos nx$. Using $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$ and $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$,

$$\int_0^\pi \sin^2 nx\,\mathrm{d}x = \tfrac{1}{2}\Big[x - \tfrac{1}{2n}\sin 2nx\Big]_0^\pi = \frac{\pi}{2},$$
$$\int_0^\pi n^2\cos^2 nx\,\mathrm{d}x = \tfrac{n^2}{2}\Big[x + \tfrac{1}{2n}\sin 2nx\Big]_0^\pi = \frac{n^2\pi}{2},$$

since $\sin 2n\pi = 0$ for an integer n. As $n \geq 1$, $n^2 \geq 1$, so $\frac{\pi}{2} \leq \frac{n^2\pi}{2}$ and $(*)$ holds (with equality exactly when $n = 1$).

Counterexamples. Take $f(x) = \pi - x$, which has $f(\pi) = 0$ but $f(0) = \pi \neq 0$. Then $f'(x) = -1$ and

$$\int_0^\pi (\pi-x)^2\,\mathrm{d}x = \Big[-\tfrac{1}{3}(\pi-x)^3\Big]_0^\pi = \frac{\pi^3}{3}, \qquad \int_0^\pi 1\,\mathrm{d}x = \pi.$$

Since $\pi^2 > 3$, we have $\frac{\pi^3}{3} > \pi$, so $(*)$ fails. Similarly $f(x) = x$ has $f(0) = 0$ but $f(\pi) = \pi \neq 0$, and gives the same two integrals $\frac{\pi^3}{3} > \pi$, so $(*)$ fails again. So neither end condition can be dropped, even when the other one holds.

Part (ii)

The quadratic. We need $f(0) = c = 0$ and $f(\pi) = a\pi^2 + b\pi = 0$, so $b = -a\pi$ and $f(x) = ax(x - \pi)$. Taking $a = 0$ would give $f \equiv 0$ and the useless inequality $0 \leq 0$, so take $a \neq 0$. Both sides of $(*)$ are multiplied by a^2 when f is multiplied by a, so we may as well take $a = -1$:

$$f(x) = x(\pi - x), \qquad f'(x) = \pi - 2x.$$

Substituting $x = \frac{\pi}{2} + u$ makes both integrals symmetric: $f = \frac{\pi^2}{4} - u^2$ and $f' = -2u$, with u running over $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Writing $h = \frac{\pi}{2}$,

$$\int_0^\pi f^2 \,\mathrm{d}x = \int_{-h}^{h} \left(h^2 - u^2\right)^2 \mathrm{d}u = 2\left(h^5 - \tfrac{2}{3}h^5 + \tfrac{1}{5}h^5\right) = \frac{16h^5}{15} = \frac{\pi^5}{30},$$
$$\int_0^\pi f'^2 \,\mathrm{d}x = \int_{-h}^{h} 4u^2 \,\mathrm{d}u = \frac{8h^3}{3} = \frac{\pi^3}{3}.$$

So $(*)$ gives $\dfrac{\pi^5}{30} \leq \dfrac{\pi^3}{3}$, and dividing by the positive number $\frac{\pi^3}{30}$,

$$\pi^2 \leq 10,$$

as required.

The trigonometric function. Now $f(0) = q + r$ and $f(\pi) = p \sin\frac{\pi}{2} + q\cos\frac{\pi}{2} + r = p + r$. Both vanish exactly when $p = q = -r$. Again $r = 0$ gives $f \equiv 0$, and scaling doesn't matter, so take $r = -1$:

$$f(x) = \sin\tfrac{1}{2}x + \cos\tfrac{1}{2}x - 1, \qquad f'(x) = \tfrac{1}{2}\left(\cos\tfrac{1}{2}x - \sin\tfrac{1}{2}x\right).$$

? **Ask yourself** How do we square f without a mess of cross terms?

Group it as $(s + c) - 1$ with $s = \sin\frac{1}{2}x$, $c = \cos\frac{1}{2}x$. Then $(s + c)^2 = 1 + 2sc = 1 + \sin x$, so

$$f^2 = (s + c)^2 - 2(s + c) + 1 = 2 + \sin x - 2\sin\tfrac{1}{2}x - 2\cos\tfrac{1}{2}x.$$

Using $\int_0^\pi \sin x \,\mathrm{d}x = 2$, $\int_0^\pi \sin\frac{1}{2}x \,\mathrm{d}x = \left[-2\cos\frac{1}{2}x\right]_0^\pi = 2$ and $\int_0^\pi \cos\frac{1}{2}x \,\mathrm{d}x = \left[2\sin\frac{1}{2}x\right]_0^\pi = 2$,

$$\int_0^\pi f^2 \,\mathrm{d}x = 2\pi + 2 - 4 - 4 = 2\pi - 6.$$

In the same way $(c - s)^2 = 1 - \sin x$, so

$$\int_0^\pi f'^2 \,\mathrm{d}x = \frac{1}{4}\int_0^\pi (1 - \sin x) \,\mathrm{d}x = \frac{\pi - 2}{4}.$$

Now $(*)$ says $2\pi - 6 \leq \frac{1}{4}(\pi - 2)$, that is $8\pi - 24 \leq \pi - 2$, so

$$7\pi \leq 22, \qquad \text{i.e.} \qquad \pi \leq \frac{22}{7}.$$

Comparing. Squaring (both sides are positive), the second inequality gives

$$\pi^2 \leq \frac{484}{49} < \frac{490}{49} = 10.$$

So the trigonometric choice gives the smaller upper bound, $\pi^2 \leq \frac{484}{49} \approx 9.878$, and hence the better estimate. Both are upper bounds, so a smaller one is closer to the truth.

Checks

- The true value is $\pi^2 \approx 9.8696$, below both 10 and $\frac{484}{49} \approx 9.8776$, as it must be.
- Numerically, for the trigonometric f the two sides of $(*)$ are $2\pi - 6 \approx 0.2832$ and $\frac{1}{4}(\pi - 2) \approx 0.2854$: very close, which is why the bound $\pi \leq \frac{22}{7}$ is so sharp. For the quadratic they are $\frac{\pi^5}{30} \approx 10.20$ and $\frac{\pi^3}{3} \approx 10.34$.
- In part (i), $n = 1$ gives equality, so $\sin x$ is the extreme case of $(*)$. The trigonometric f is close in shape to $\sin x$, while $x(\pi - x)$ is a little less so, which explains which bound is better.

Common slips

- Choosing a counterexample such as $f = 1$ that breaks *both* end conditions. It is valid, but $\pi - x$ and x show more: each condition is needed on its own.
- Forgetting $\int_0^\pi \sin x \, \mathrm{d}x = 2$ (not 0) when integrating f^2 in the trigonometric case.
- Saying 10 is the better estimate because it is "rounder". Both are upper bounds, so the smaller one wins.

Beyond STEP

Inequality $(*)$ is Wirtinger's inequality, and the best constant 1 comes from $\sin x$: it is the smallest eigenvalue of $-f'' = \lambda f$ with $f(0) = f(\pi) = 0$. Testing $(*)$ on chosen functions is the Rayleigh–Ritz method, which always gives upper bounds for that eigenvalue and underlies how engineers estimate vibration frequencies. Fourier series give a slick proof of $(*)$ itself. For the analysis behind this, see Abbott, *Understanding Analysis*; for the eigenvalue side, Boyce and DiPrima, *Elementary Differential Equations*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.